



AI Playbook and Toolkit for Public Health Departments

AI Incident Response for Public Health

GOV 120

This module prepares public health agencies to recognize, document, escalate, and respond to AI-related incidents. An AI incident may involve inaccurate outputs, inappropriate disclosure, harmful recommendations, biased results, unsafe automation, public communication errors, vendor failures, security events, or loss of public trust. Incident response is part of AI governance because AI systems require ongoing oversight after initial approval. The module is designed for national use and does not assume a specific agency incident response system. Learners should adapt the concepts to their own privacy, cybersecurity, legal, communications, emergency response, records, and governance procedures. The module emphasizes that AI incident response should be planned before deployment rather than invented during a crisis.

Level	Intermediate
Primary track	Governance and Security
Appears in tracks	governance-security

Learning Objectives

- Define an AI incident and distinguish it from a routine output error.
- Describe the steps in AI incident triage, containment, documentation, corrective action, and lessons learned.
- Identify which reviewers may be needed for different AI incident types.
- Create an AI incident response record for a realistic public health scenario.
- Explain why near misses should be documented and reviewed.

Module Overview

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The module is designed for national use and does not assume a specific agency incident response system. Learners should adapt the concepts to their own privacy, cybersecurity, legal, communications, emergency response, records, and governance procedures. The module emphasizes that AI incident response should be planned before deployment rather than invented during a crisis.

Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace legal, privacy, procurement, civil rights, records, cybersecurity, Tribal consultation, labor relations, human resources, or agency policy review. Learners should confirm applicable requirements in their own jurisdiction and organization before implementing AI-supported workflows.

Why This Topic Matters for Public Health AI

Public health agencies make decisions that affect communities, services, resources, public communication, and emergency response. When an AI-supported workflow fails, the consequences may include misinformation, missed signals, inappropriate prioritization, disclosure of sensitive information, reduced trust, or harm to populations already experiencing inequities. These risks are more difficult to manage if the agency has no defined path for reporting and responding to AI concerns.

AI incidents can also be ambiguous. A harmful output may result from poor data quality, an inappropriate prompt, a model limitation, a vendor change, weak human review, or a workflow design problem. A structured response process helps the agency determine what happened, what immediate safeguards are needed, who must be notified, and what corrective action should follow.

Public Health Example

A local health department uses an approved AI-supported tool to draft internal summaries of disease surveillance signals. During a busy week, staff notice that one summary incorrectly describes a cluster as occurring in a school setting when the underlying data do not support that conclusion. The draft was not released publicly, but it was shared internally and could have affected follow-up priorities.

The agency treats this as a near miss. Staff document the error, identify who received the summary, review whether any decisions were affected, pause use of the summary template, and notify the program owner and AI governance contact. The corrective action includes updating the prompt, requiring a data-source verification step, and adding the error type to ongoing monitoring.

Technical, Programmatic, and Operational Deep Dive

AI incident response should connect to existing agency processes rather than operate in isolation. Privacy incidents, cybersecurity events, communication errors, records issues, procurement disputes, program risks, and emergency operations may already have reporting pathways. AI governance should define when those pathways are triggered and how the AI committee or governance body is informed.

Incident response should include immediate containment, fact-finding, impact assessment, notification decisions, corrective action, and lessons learned. Containment may mean pausing a tool, disabling a feature, restricting access, removing public content, or switching to a manual workflow. Fact-finding should preserve relevant prompts, outputs, logs, data sources, versions, reviewer notes, and decisions.

Leadership should decide in advance what types of incidents require escalation. High-priority examples include possible exposure of protected or confidential data, public-facing misinformation, biased or discriminatory impact, security compromise, automated action without authorization, vendor model changes affecting performance, or repeated errors that suggest the tool is not fit for use. Escalation thresholds should be clear enough for frontline staff to act quickly.

Risks, Failure Modes, and Guardrails

A common failure mode is blaming the user without examining the system. AI incidents often reflect gaps in workflow design, tool configuration, training, data quality, monitoring, vendor controls, or governance conditions. Guardrails include root cause analysis, nonpunitive reporting, documentation, and attention to both technical and human factors.

Another failure mode is continuing to use an AI tool after serious concerns are identified because the workflow has become dependent on it. Agencies should define pause criteria before deployment so staff know when a tool must be stopped, limited, or reverted to manual review. A pause should be treated as a responsible safety action, not as a failure of innovation.

A third risk is failing to communicate clearly after an AI-related error. Some incidents may require internal communication, partner notification, public correction, legal review, or leadership messaging. Guardrails include pre-defined communication roles, plain-language explanations, documentation of what is known and unknown, and transparency about corrective action when disclosure is appropriate.

Application to AI-Supported Workflows

Every AI-supported workflow should include an incident reporting step. Staff should know what to report, where to report it, who responds, and what information should be preserved. Reporting should include errors, near misses, unexpected outputs, user concerns, public complaints, equity concerns, security concerns, and vendor behavior changes.

AI incident response should also feed continuous improvement. Incident trends may reveal that a use case needs additional training, stronger human review, a revised risk classification, better monitoring, or retirement. Governance bodies should review incident patterns as part of lifecycle oversight.

Reflection Questions

What AI-related events would require immediate escalation in your agency?

Who would need to be involved if an AI-supported public message contained an error?

What evidence should be preserved when an AI incident is reported?

Practical Exercise

Create an AI incident response record for a realistic scenario. Choose an incident type such as inaccurate public-facing content, possible sensitive data disclosure, biased prioritization output, unsafe automation, vendor system change, or repeated surveillance summary errors. Use fictional or generalized information only.

Your record should include incident description, date discovered, system or workflow involved, affected users or populations, data involved, immediate containment action, reviewers needed, severity, corrective action, communication needs, and monitoring follow-up.

Example: An AI-assisted communications draft incorrectly states that a boil water advisory applies to all county residents when it applies only to a specific water district. The incident record identifies the draft source, who reviewed it, whether it was published, who needs notification, how the message will be corrected, and what review step will be added before future advisories are released.

Expected Artifact or Evidence

Completed AI Incident Response Record

Corrective Action and Follow-Up Plan

Knowledge Check

- 1. What is the most important first step after completing the AI Incident Response for Public Health module?
- 2. Which practice best supports responsible AI use in public health?
- 3. Which statement best reflects the risk or guardrail for this module: A common failure mode is blaming the user without examining the system. AI incidents often reflect gaps in workflow design, tool configuration, training, data quality, monitoring, vendor controls, or governance conditions. Guardrails include root cause analysis, nonpunitive reporting, documentation, and attention to both technical and human factors.
- Another failure mode is continuing to use an AI tool after serious concerns are identified because the workflow has become dependent on it. Agencies should define pause criteria before deployment so staff know when a tool must be stopped, limited, or reverted to manual review. A pause should be treated as a responsible safety action, not as a failure of innovation.
- A third risk is failing to communicate clearly after an AI-related error. Some incidents may require internal communication, partner notification, public correction, legal review, or leadership messaging. Guardrails include pre-defined communication roles, plain-language explanations, documentation of what is known and unknown, and transparency about corrective action when disclosure is appropriate.
- 4. When should a staff member escalate an AI-related concern?
- 5. What counts as stronger completion evidence for a training module?

References and Resources

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